Paul Okewunmi

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Summary

AI engineer and researcher with 4+ years building and shipping LLM systems, from conversational agents and evaluation frameworks in production to multilingual NLP at Meta. Experienced in fine-tuning and post-training, RAG, and MLOps, with published work on LLM robustness and AI safety (ACL 2025–26).

Education

- **Carnegie Mellon University Africa** — Kigali, Rwanda
  M.S. in Engineering Artificial Intelligence — *Aug 2026 – May 2028 (expected)*
  - **Coursework:** Maths Foundation for ML (04-650), Agentic AI (04-801-W2), Data & Inference (18-785).
- **Obafemi Awolowo University** — Ile-Ife, Nigeria
  B.Sc. Electronic and Electrical Engineering; GPA 4.57/5.00 (First Class Honours) — *Apr 2018 – Sept 2024*
  - **Teaching Assistant**, EEE503 Control Systems Engineering (Nov 2024 – Mar 2025): state estimation (Kalman filters), model predictive control.

Publications & Posters

- *UbuntuGuard: A Culturally-Grounded Policy Benchmark for Equitable AI Safety in African Languages.* **ACL Findings 2026**. [paper]
- *Evaluating Robustness of LLMs to Typographical Noise in Yorùbá QA*. AfricaNLP Workshop @ ACL 2025. **Best Paper Award.** [paper]
- *Afri-Aya: Giving Sight to African LLMs.* [dataset]
- *Adaptive-Size Fiducial Markers with Dual-Stage Kalman Filtering for Precision Rotary-Wing UAV Landing.* Nigerian Journal of Technological Development, 2025. [paper]
- *Accelerometry-Based Improvement of F450 Quadcopter Flight Control.* IEEE, 2024. [paper]
- *Unmasking 'Fake News': A Pipeline for Fact-Checking Radio Shows via AI-Driven Claim Extraction.* Deep Learning Indaba, 2023. [poster]
- *Drone-Based Monitoring System for Large-Scale Farming.* AI Bootcamp Nigeria, 2021. **Best Poster Award.** [poster]

Industry Experience

- **5.Y Inc.** — Singapore (Remote)
  Founding AI Engineer — *Dec 2023 – Jul 2025*
  - Led development of GLUCOSE, a conversational AI agent with custom tools for customer engagement.
  - Built an agent evaluation framework that simulates user personas to test reliability before and after deployment.
  - Set up production tracing and monitoring for agents with LangSmith, improving observability and reliability.
- **Meta (MLH Fellowship)** — Remote
  Software Engineering Intern — *Sept 2023 – Dec 2023*
  - Contributed to the public release of Meta's laser-encoders library on PyPI for multilingual sentence embeddings.
  - Built a production-ready Docker image with a REST interface for sentence embeddings in 200+ languages.
  - Developed interactive demos of the cross-lingual sentence clustering capabilities of LASER.
- **Centre for Journalism Innovation and Development** — Abuja, Nigeria
  MLOps Engineer — *Aug 2022 – Aug 2023*
  - Led the initial AI development for Dubawa Audio, an AI-powered fact-checking platform for radio.
  - Fine-tuned BERT-based models for automated claim detection, reducing fact-verification time by about 80%.
  - Built end-to-end pipelines for audio collection, speech-to-text, model training, and deployment.
- **AirQo** — Kampala, Uganda
  ML Engineering Intern — *Apr 2022 – Jun 2022*
  - Deployed fault-detection models for air quality sensors across Uganda, reducing sensor downtime.
  - Trained a multi-label classifier for sensor fault categories with over 90% accuracy.

Research Experience

- **Cohere Labs Community** — Remote
  Yorùbá Lead, Afri-Aya — *May 2025 – Present*
  - Curating culturally grounded African-language datasets for fine-tuning language models.
- **ML Collective** — Remote
  Project Lead — *Feb 2025 – Present*
  - Leading a research project applying model-cascading techniques to multimodal tasks.

Selected Projects

- **Smol-Experiments: How Much Research Is Possible on a Single T4 GPU?** (GitHub)
  - Implemented the SmolLM-135M architecture from scratch as a testbed for modern training and alignment methods.
  - Implemented SFT, DPO, and CPO to train grammatical-error-correction instruction following on CoEdIT.
  - Converted the dense model into a Mixture-of-Experts (SmolMoE) via sparse upcycling; trained routing on Nemotron Post-Training data, reaching 80%+ expert-routing accuracy across code, math, and chat.
- **SabiEssay: Rubric-Based Grading of WAEC-Style Essays with Small LLMs** (Dataset, Model)
  - Built a synthetic data pipeline generating WAEC-style essays graded against the WAEC marking guide.
  - Released SabiEssay v0.1, a dataset of 194 rubric-scored essays with examiner-style feedback, for fine-tuning small LLMs as automated essay graders.
  - Fine-tuned a 2B model that runs on basic consumer PC for on-device essay grading and feedback.
- **Jumia Lens: Visual Product Search Engine** (GitHub)
  - Built an end-to-end image retrieval system for e-commerce products using metric learning, trained on 3,650+ scraped product images.
  - Served embeddings and metadata through Pinecone for low-latency vector search.
- **Òmòwé.ai: Multilingual Learning Assistant** (GitHub)
  - Built a RAG system for searching study materials in 100+ languages, with summarization, Q&A generation, and paraphrasing. 3rd place, Cohere Multilingual AI Hackathon.
- **Code4me.AI: AI-Powered VS Code Extension** (GitHub)
  - Led a team of 3 to build a VS Code extension for code generation, auto-debugging, unit-test generation, and documentation from natural language.

Honors & Awards

- Emerging Leader in AI, Black in AI 2025
- 1st Place, Cohere Labs Expedition Aya Challenge [link] 2025
- 1st Place, Tunga Africa Generative AI Hackathon [link] 2023
- 3rd Place, InstaDeep AI Hackathon, Deep Learning Indaba [link] 2023
- 3rd Place, Cohere Multilingual AI Hackathon [link] 2023
- 1st Place, Blackathon [link] 2022
- Great Ife Alumni Scholarship 2022
- Federal Government Scholarship 2021
- Mr. Algorithm Award, Data Science Nigeria [link] 2020

Invited Talks

- *When Typos Break AI: Testing the Limits of LLM Robustness in Low-Resource Languages.* Cohere Labs Community. [video]
- *Tools for the Future: Python Frameworks and Tools in the GenAI Era.* PyCon Africa 2024. [link]
- *A Picture Is Worth a Thousand Words: Building a Visual Search Engine.* PyCon Nigeria 2024.
- *Prompt Engineering 101: Building Prompt-Based Applications.* DataFest Africa 2023. [video]

Service & Leadership

- Campus Lead, Data Science Nigeria (OAU chapter): ran ML workshops and study groups 2019 – 2024
- Technical Director, IEEE Student Branch, Obafemi Awolowo University 2021 – 2022

Skills

Languages: Python, JavaScript, MATLAB, SQL
ML Frameworks: PyTorch, Hugging Face, vLLM, SGLang, LangChain, LlamaIndex, OpenCV, MLflow
Infrastructure: Docker, Git, AWS, GCP, PostgreSQL, MongoDB, Neo4j, Pinecone, Prefect